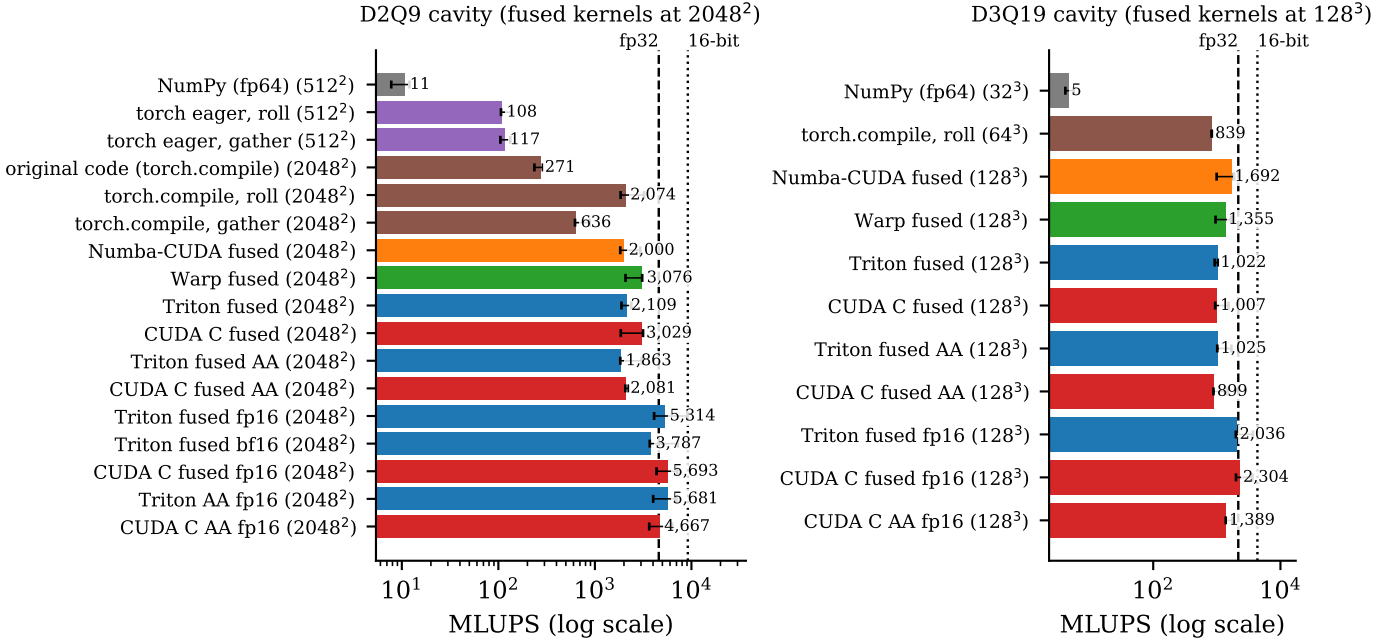
D2Q9 cavity (fused kernels at 2048²)
fp32
16-bit
NumPy (fp64) (512²)
torch eager, roll (512²)
torch eager, gather (512²)
original code (torch.compile) (2048²)
torch.compile, roll (2048²)
torch.compile, gather (2048²)
Numba-CUDA fused (2048²)
Warp fused (2048²)
Triton fused (2048²)
CUDA C fused (2048²)
Triton fused AA (2048²)
CUDA C fused AA (2048²)
Triton fused fp16 (2048²)
Triton fused bf16 (2048²)
CUDA C fused fp16 (2048²)
Triton AA fp16 (2048²)
CUDA C AA fp16 (2048²)
11
108
117
271
2,074
636
2,000
3,076
2,109
3,029
1,863
2,081
5,314
3,787
5,693
5,681
4,667
10¹
10²
10³
10⁴
MLUPS (log scale)
D3Q19 cavity (fused kernels at 128³)
fp32
16-bit
NumPy (fp64) (32³)
torch.compile, roll (64³)
Numba-CUDA fused (128³)
Warp fused (128³)
Triton fused (128³)
CUDA C fused (128³)
Triton fused AA (128³)
CUDA C fused AA (128³)
Triton fused fp16 (128³)
CUDA C fused fp16 (128³)
CUDA C AA fp16 (128³)
5
839
1,692
1,355
1,022
1,007
1,025
899
2,036
2,304
1,389
10²
10⁴
MLUPS (log scale)